

Lecture 3

learning and regularization

iterative methods

Stéphane Canu and Cheng Soon Ong
`stephane.canu@insa-rouen.fr`
`asi.insa-rouen.fr/~scanu`

RSISE ANU - NICTA, Canberra - INSA, Rouen
RSISE, SML group - Laboratoire PSI

Appetizer...adjoint operator

- let $\mathcal{H}$ be a Hilbert space with dot product $\langle f, g \rangle_{\mathcal{H}}$
- let T be a linear operator

$$\begin{array}{ll} T : \mathcal{H} \longrightarrow \mathbb{R}^m & T^* : \mathbb{R}^m \longrightarrow \mathcal{H} \\ f \longmapsto Tf & \alpha \longmapsto T^*\alpha \end{array}$$

- define the adjoint T^* when it exists with

$$\langle Tf, \alpha \rangle_{\mathbb{R}^m} = \langle f, T^*\alpha \rangle_{\mathcal{H}}$$

- T^* is also linear
- because TT^* is also linear, there exists a matrix K such that :

$$\begin{array}{ll} TT^* : \mathbb{R}^m \longrightarrow \mathbb{R}^m & \\ \alpha \longmapsto TT^*\alpha = K\alpha & \end{array}$$

Previously on functional learning and regularization

- interpolation : given a sample $S_m = \{x_i \in \Omega, y_i \in \mathbb{R}, \quad i = 1, m\}$
find $f \in \mathcal{H}$ such that $Tf = \mathbf{y}$ i.e. $f(x_i) = y_i, \quad i = 1, m$
- define $\mathcal{H}$ hypothesis space
 - $\mathcal{H} \subset \mathbb{R}^\Omega$ pointwise define functions
 - T is continuous $\Leftrightarrow$ there exists a kernel $\kappa(x, x')$
 - $\forall f \in \mathcal{H}, \forall x_i \in \mathbb{R}, \quad f(x_i) = \langle \kappa(x, \cdot), f(\cdot) \rangle_{\mathcal{H}}$ reproducing property
- $\mathcal{H}$ is a r.h.s. $\Leftrightarrow \kappa(x, x') \Leftrightarrow$ operator $S : L^2 \longrightarrow \mathbb{R}^\Omega$
- $Tf = \mathbf{y} \Leftrightarrow K\alpha = y$
- regularize with iterative methods using early stopping

Road map

- ill posed problem, regularization and Learning
 1. splines
 2. well posed and ill posed problem
 3. regularization methods
 4. filter functions
- **Iterative algorithm and regularization**
 - interpolation
 - illustration
 - conjugate gradient type methods
- Semi convergence - the proof

The interpolation problem

- inputs : $\mathbf{x} \in \mathbb{R}^d$, output : $y \in \mathbb{R}$
- sample : $S_m = \{\mathbf{x}_1, y_1, \dots, \mathbf{x}_m, y_m\}$

Definition (Evaluation functional)

$$T : \mathcal{H} \longrightarrow \mathbb{R}^m$$

$$f \longmapsto Tf = (f(\mathbf{x}_1), \dots, f(\mathbf{x}_i), \dots, f(\mathbf{x}_m))^{\top}$$

Definition (Interpolation problem in $\mathcal{H}$)

find $f \in \mathcal{H}$ such that $Tf = \mathbf{y}$

To be well posed the interpolation problems needs

- choose $\mathcal{H}$ such that T is well defined and continuous
- add a constraint to get unicity

Definition (Minimum norm interpolation problem)

Let $\mathcal{H} \subset \mathbb{R}^{\Omega}$ in which the evaluation functional T is continuous ($\mathcal{H}$ is a r.k. space)

$$\min_{f \in \mathcal{H}} \|f\|^2 \text{ such that } Tf = \mathbf{y}$$

let's solve the minimum norm interpolation problem (M.N.I.P.)

Representer theorem for the M.N.I.P.

Representer theorem

$$\begin{cases} \min_{f \in \mathcal{H}} \|f\|^2 \\ \text{such that } Tf = \mathbf{y} \end{cases} \Leftrightarrow \begin{cases} f = T^* \alpha \\ \text{solve } K\alpha = \mathbf{y} \end{cases} \quad K_{ij} = \kappa(x_i, x_j)$$

evaluation operator and its adjoint

$$T: H \longrightarrow \mathbb{R}^m$$

$$f \longmapsto Tf = (f(x_1), \dots, f(x_m))^\top$$

$$T^*: \mathbb{R}^m \longrightarrow H$$

$$\alpha \longmapsto T^* \alpha = \sum_{i=1}^m \alpha_i \kappa(\cdot, x_i)$$

build the adjoint

$$\begin{aligned} \langle Tf, \alpha \rangle_{\mathbb{R}^m} &= \sum_{i=1}^m f(x_i) \alpha_i \\ &= \sum_{i=1}^m \langle f, \kappa(\cdot, x_i) \rangle_{\mathcal{H}} \alpha_i \\ &= \langle f, \underbrace{\sum_{i=1}^m \alpha_i \kappa(\cdot, x_i)}_{T^* \alpha} \rangle_{\mathcal{H}} \end{aligned}$$

$$TT^*: \mathbb{R}^m \longrightarrow \mathbb{R}^m$$

$$\begin{aligned} \alpha \longmapsto TT^* \alpha &= \sum_{j=1}^m \alpha_j \kappa(x_j, x_i) \\ &= K\alpha \end{aligned}$$

the eigen structure. Assume K is full rank - $K\phi_i = \mu_i \phi_i$

the use of the eigen structure

- Assume K is full rank - $K\phi_i = \mu_i\phi_i$
- functions $T^*\phi_i, i = 1, m$ generates F_m a linear subspace of $\mathcal{H}$, such that $\mathcal{H} = F_m \oplus G$ note that $\forall g \in G, Tg = 0$

$$\begin{aligned}\forall f \in \mathcal{H}, \quad f(.) &= \sum_{i=1}^m \beta_i T^* \phi_i + g(.) \\ &= \sum_{i=1}^m \beta_i \left(\sum_{j=1}^m \phi_{ij} \kappa(., x_j) \right) + g(.) \\ &= \sum_{j=1}^m \left(\sum_{i=1}^m \beta_i \phi_{ij} \right) \kappa(., x_j) + g(.) \\ &= \sum_{j=1}^m \alpha_j \kappa(., x_j) + g(.) = T^* \alpha + g(.)\end{aligned}$$

- $Tf = \mathbf{y} \Leftrightarrow TT^* \alpha = K\alpha = \mathbf{y}$
- because we are looking for the minimum norm solution, $g = 0$

summarize the M.N.I.P.

$$\begin{cases} \min_{f \in \mathcal{H}} \|f\|^2 \\ \text{such that } Tf = y \end{cases}$$

- chose the kernel $\kappa(.,.)$
- build the gram matrix K , $(K_{ij} = \kappa(x_i, x_j))$

- solve $K\alpha = y$

- $f = T^*\alpha = \sum_{j=1}^m \alpha_j \kappa(., x_j)$

- cost $\mathcal{O}(m^3)$

- what if K is singular?

regularize more

- what is we want to approximate instead of interpolate?

regularize more

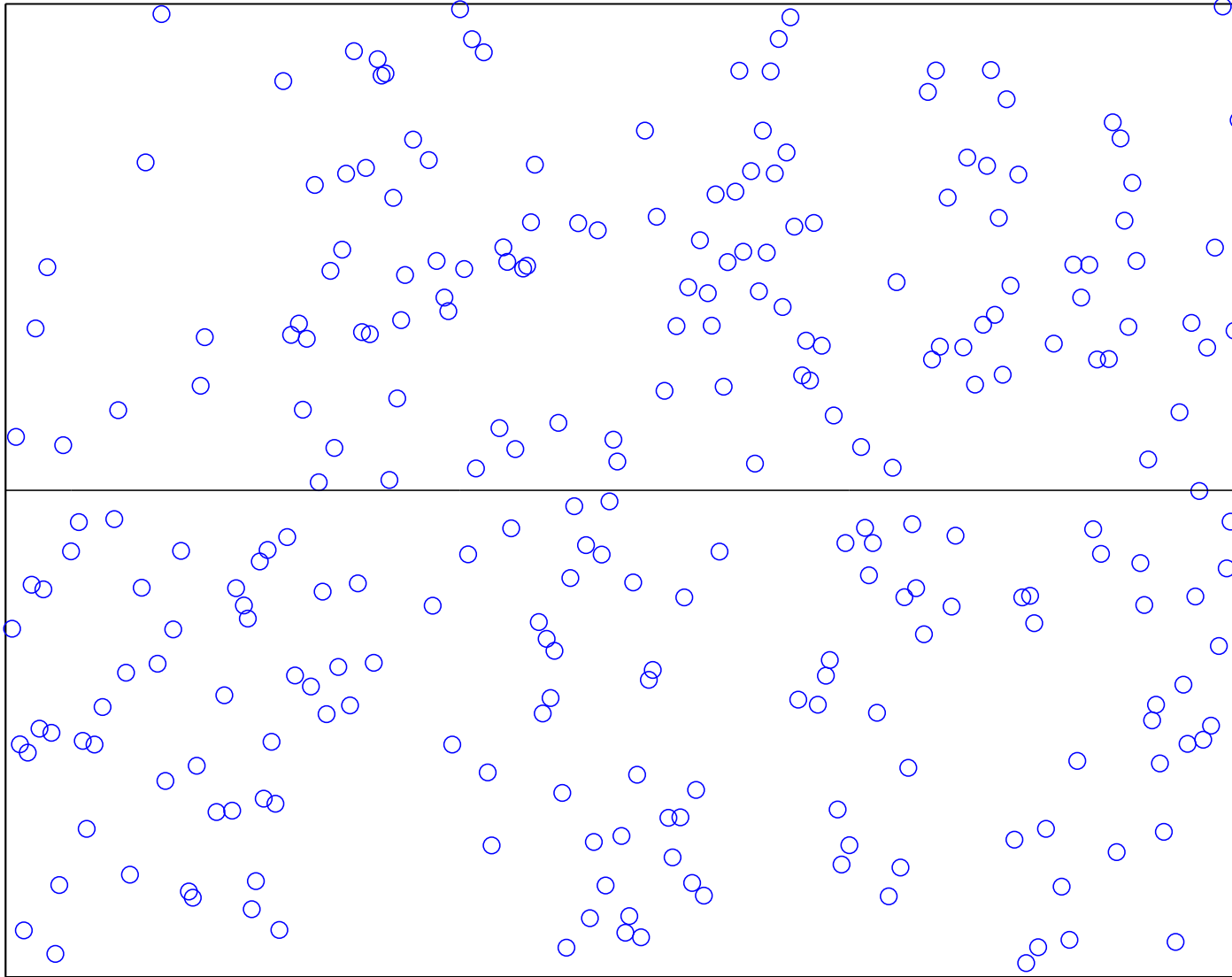
- what if there is no norm?

use orthogonality

- project 0 onto the interpolating set $I_{\mathcal{H}} = \{f \in \mathcal{H} \text{ such that } Tf = y\}$,

- project $I_{\mathcal{H}}$ onto $F_m = \text{span}\{\kappa(., x_1), \dots, \kappa(., x_m)\}$

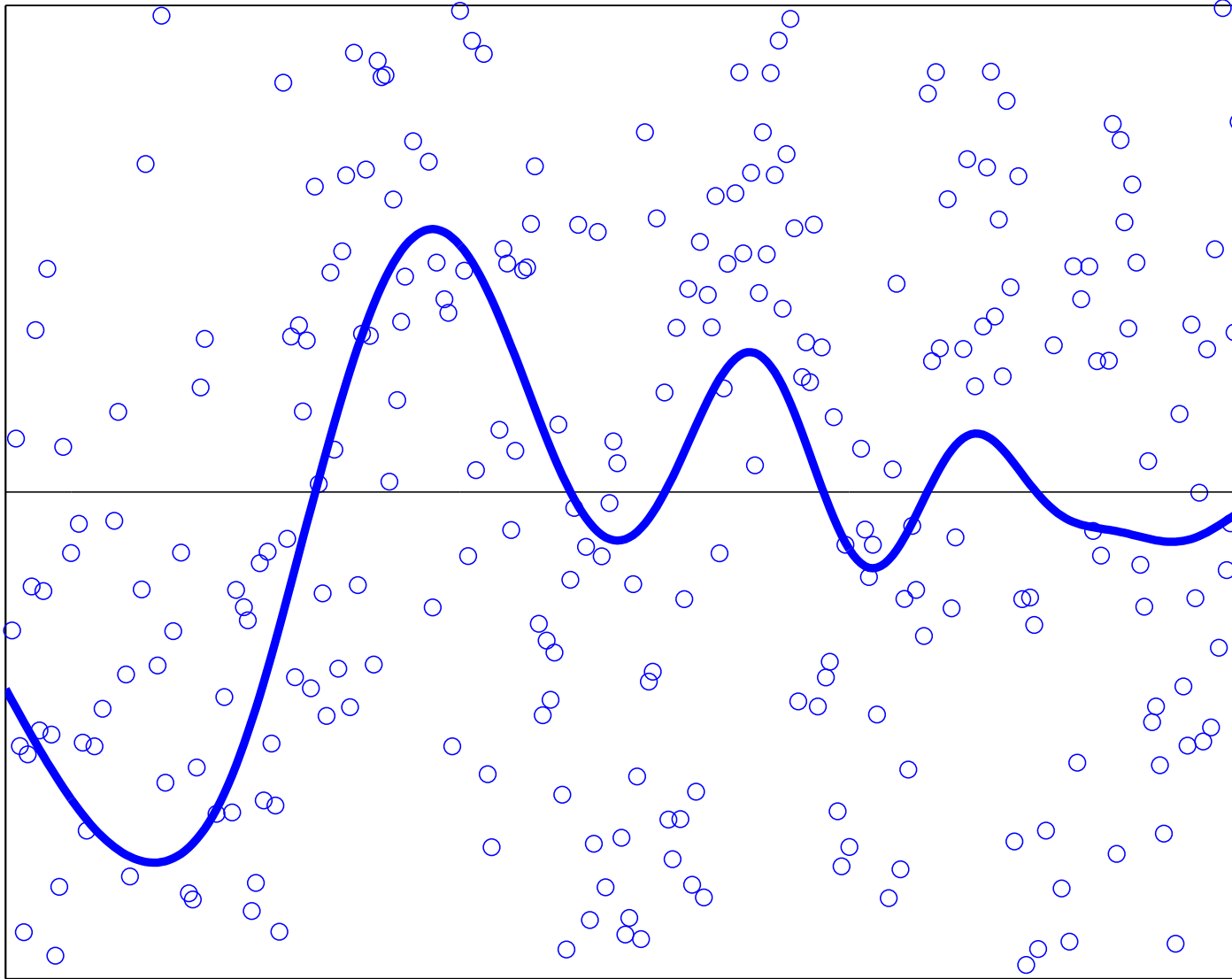
iterative method at work...



○ the data

$$x_i, y_i, i = 1, 500$$

iterative method at work...

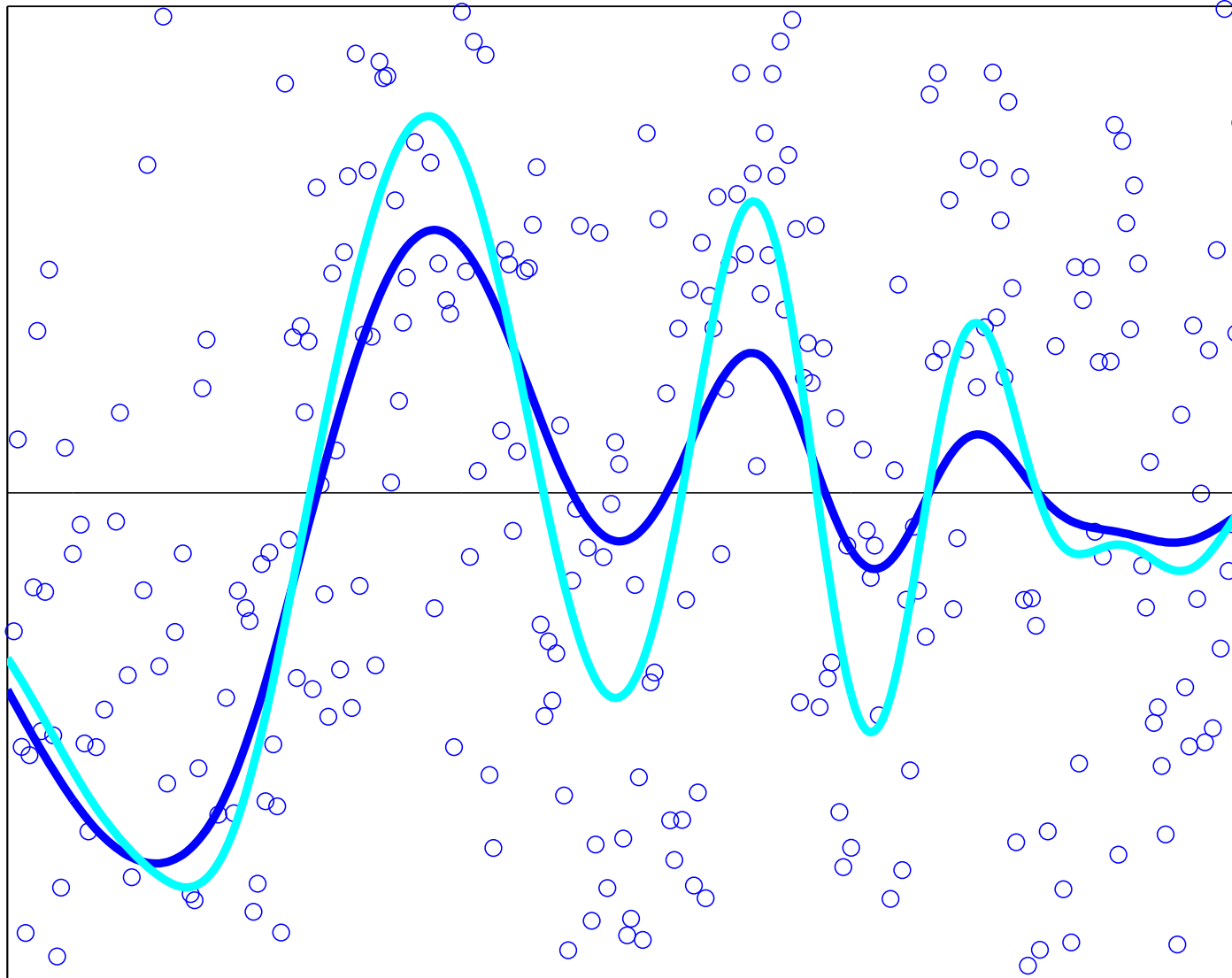


○ the data

$$x_i, y_i, i = 1, 500$$

– f_1 approximation
after 1 iteration

iterative method at work...



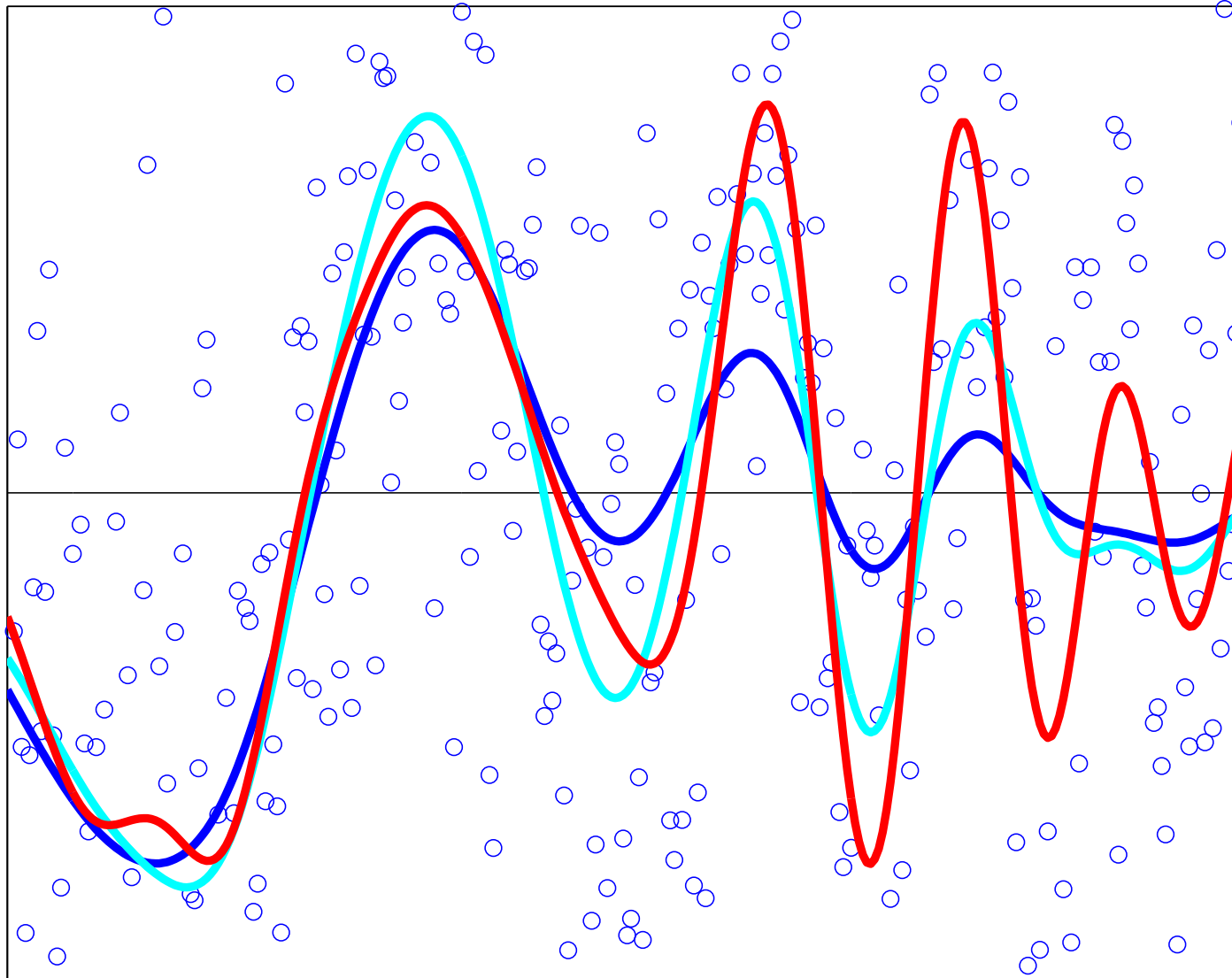
○ the data

$$x_i, y_i, i = 1, 500$$

– f_1 approximation
after 1 iteration

– f_2 approximation
after 2 iterations

iterative method at work...



○ the data

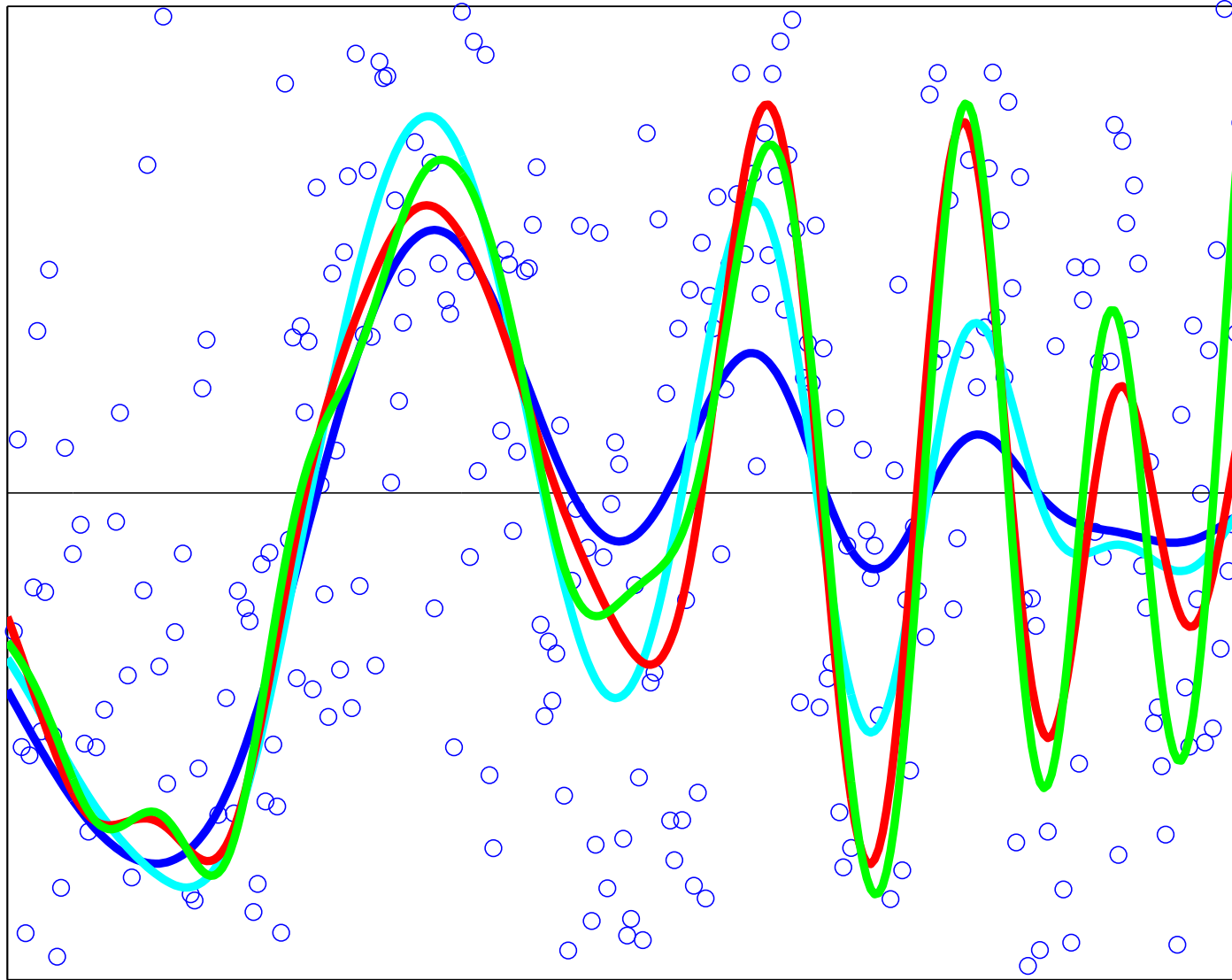
$$x_i, y_i, i = 1, 500$$

– f_1 approximation
after 1 iteration

– f_2 approximation
after 2 iterations

– f_5 approximation
after 5 iterations

iterative method at work...

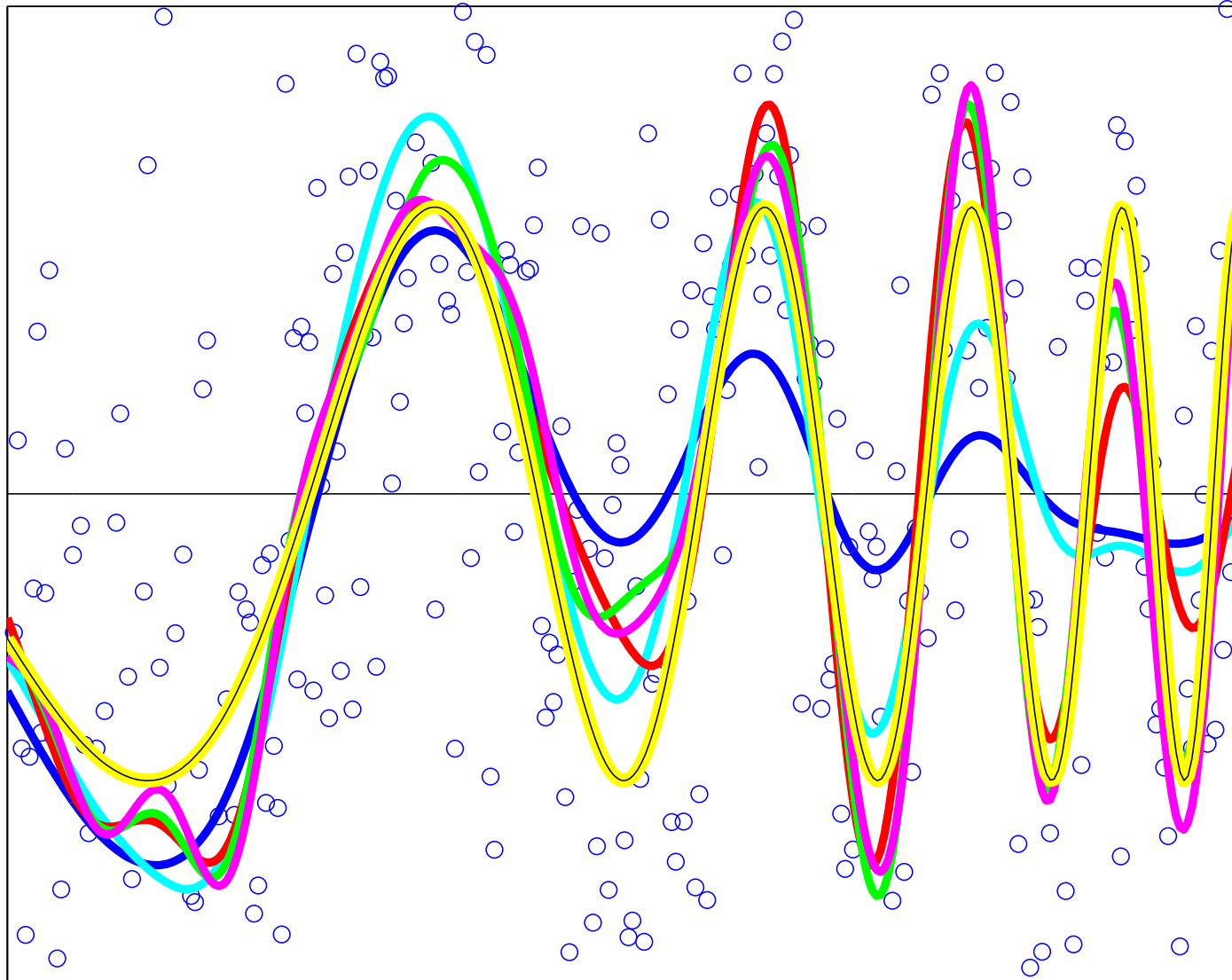


○ the data

$$x_i, y_i, i = 1, 500$$

- f_1 approximation after 1 iteration
- f_2 approximation after 2 iterations
- f_5 approximation after 5 iterations
- f_9 approximation after 9 iterations

iterative method at work...



○ the data

$$x_i, y_i, i = 1, 500$$

– f_1 approximation
after 1 iteration

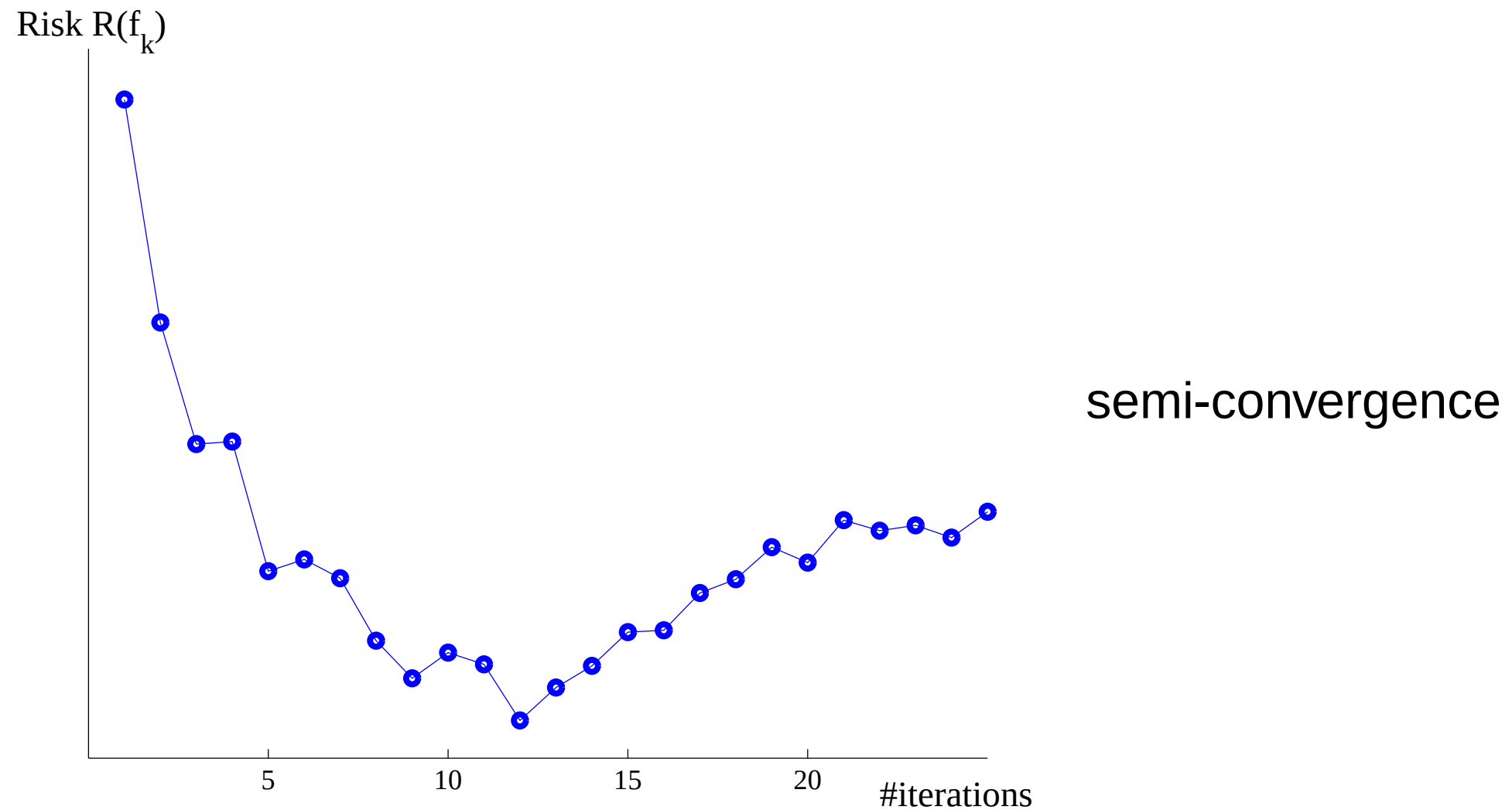
– f_2 approximation
after 2 iterations

– f_5 approximation
after 5 iterations

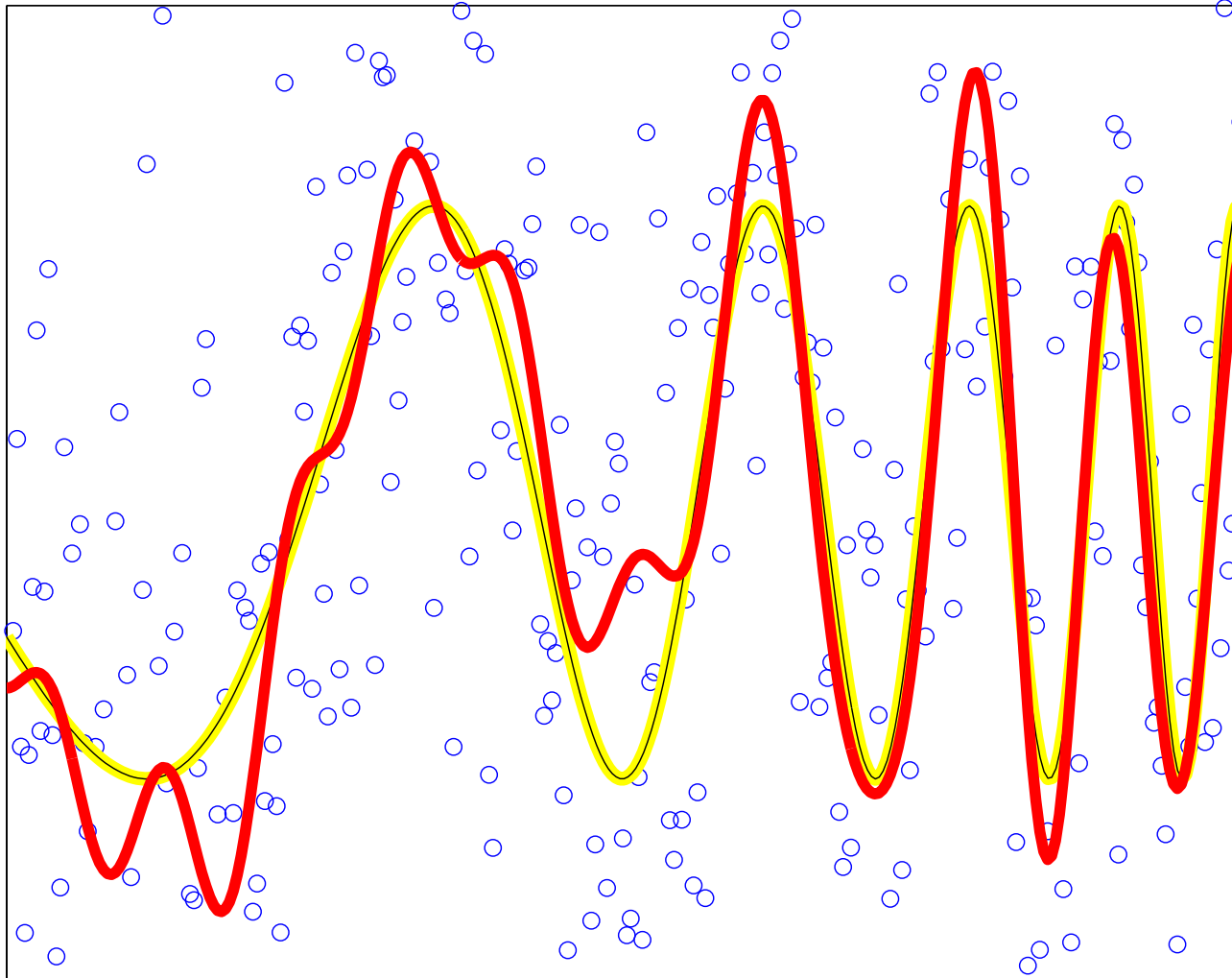
– f_9 approximation
after 9 iterations

– function $t(x)$
the target

Convergence of the iterative method at work...



overfitting of the iterative method at work...



Overfitting

conjugate gradient algorithm - step 1 decoupling

Variational formulation (assume K is symmetric positive definite)

$$K\alpha = y \quad \Leftrightarrow \quad \min_{\alpha \in \mathbb{R}^m} J(\alpha) \quad \text{with} \quad J(\alpha) = \frac{1}{2} \alpha^\top K \alpha - \alpha^\top y$$

Choose $\{\mathbf{d}_1, \dots, \mathbf{d}_m\}$ a basis in $\mathbb{R}^m$, then

$$\alpha = \sum_{k=1}^m a_k \mathbf{d}_k$$

$$\begin{aligned} J(\alpha) &= \frac{1}{2} \left(\sum_{k=1}^m a_k \mathbf{d}_k \right)^\top K \left(\sum_{j=1}^m a_j \mathbf{d}_j \right) - \left(\sum_{k=1}^m a_k \mathbf{d}_k \right)^\top y \\ &= \frac{1}{2} \sum_{k=1}^m \sum_{j=1}^m a_k a_j (\mathbf{d}_k^\top K \mathbf{d}_j) - \sum_{k=1}^m a_k (\mathbf{d}_k^\top y) \end{aligned}$$

cancel the cross terms by building a "good" basis $\mathbf{d}_k$

$$\mathbf{d}_k^\top K \mathbf{d}_j = 0, k \neq j \quad \Rightarrow \quad J(\alpha) = \frac{1}{2} \sum_{k=1}^m a_k^2 (\mathbf{d}_k^\top K \mathbf{d}_k) - \sum_{k=1}^m a_k (\mathbf{d}_k^\top y)$$

benefit - the optimal a_k are easy to compute

$$\frac{dJ(\alpha)}{da_k} = 0 \quad \Leftrightarrow \quad a_k (\mathbf{d}_k^\top K \mathbf{d}_k) - (\mathbf{d}_k^\top y) = 0$$

$$\text{The two first steps of the CG} \begin{cases} (1) & a_k = \frac{\mathbf{d}_k^\top y}{\mathbf{d}_k^\top K \mathbf{d}_k} \\ (2) & \alpha_k = \alpha_{k-1} + a_k \mathbf{d}_k \end{cases}$$

CG algorithm - step 2 computing conjugate directions

- objective : "orthogonalize" $k \neq j$, $\mathbf{d}_k^\top K \mathbf{d}_j = 0$
- gram-schmidt orthogonalization
 - find a non vanishing direction : $\mathbf{r}_k = \mathbf{y} - K \alpha_k$
 - eliminate the useless components

$$\mathbf{d}_{k+1} = \mathbf{r}_k - b_k \mathbf{d}_k - \underbrace{b_{k-1} \mathbf{d}_{k-1} \dots - b_1 \mathbf{d}_1}_{b_{k-1} = \dots = b_1 = 0}$$

- conjugate

$$\mathbf{d}_k^\top K \mathbf{d}_{k+1} = 0 \quad \Leftrightarrow \quad \mathbf{d}_k^\top K \mathbf{r}_k - b_k (\mathbf{d}_k^\top K \mathbf{d}_k)$$

the CG algorithm - initialization $\alpha_0 = 0, \Rightarrow r_0 = \mathbf{y}, \mathbf{d}_1 = \mathbf{y}$

$$(1) \quad a_k = \frac{\mathbf{d}_k^\top \mathbf{y}}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(2) \quad \alpha_k = \alpha_{k-1} + a_k \mathbf{d}_k$$

$$(3) \quad \mathbf{r}_k = \mathbf{y} - K \alpha_k$$

$$(4) \quad b_k = \frac{\mathbf{d}_k^\top K \mathbf{r}_k}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(5) \quad \mathbf{d}_{k+1} = \mathbf{r}_k + b_k \mathbf{d}_k$$

the solution
the residual

from conjugate gradient to polynomials

$$(1) \quad a_k = \frac{\mathbf{d}_k^\top \mathbf{y}}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(2) \quad \alpha_k = \alpha_{k-1} + a_k \mathbf{d}_k \quad \text{the solution}$$

$$(3) \quad \mathbf{r}_k = \mathbf{y} - K \alpha_k \quad \text{the residual}$$

$$(4) \quad b_k = \frac{\mathbf{d}_k^\top K \mathbf{r}_k}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(5) \quad \mathbf{d}_{k+1} = \mathbf{r}_k + b_k \mathbf{d}_k$$

- each iteration costs $\mathcal{O}(m^2)$
- converge in m iteration towards the solution α such that $K\alpha = \mathbf{y}$
- can be extended to non linear equations
- polynomial and subspace interpretation

$$\alpha_0 = 0$$

$$\alpha_1 = a_1 \mathbf{y}$$

$$\alpha_2 = a_1 \mathbf{y} + a_2 \left(\underbrace{\mathbf{r}_1}_{\mathbf{y} - a_1 K \mathbf{y}} + b_1 \mathbf{y} \right)$$

$$= q_1 \mathbf{y} + q_2 K \mathbf{y}$$

from conjugate gradient to polynomials

$$(1) \quad a_k = \frac{\mathbf{d}_k^\top \mathbf{y}}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(2) \quad \alpha_k = \alpha_{k-1} + a_k \mathbf{d}_k \quad \text{the solution}$$

$$(3) \quad \mathbf{r}_k = \mathbf{y} - K \alpha_k \quad \text{the residual}$$

$$(4) \quad b_k = \frac{\mathbf{d}_k^\top K \mathbf{r}_k}{\mathbf{d}_k^\top K \mathbf{d}_k}$$

$$(5) \quad \mathbf{d}_{k+1} = \mathbf{r}_k + b_k \mathbf{d}_k$$

- each iteration costs $\mathcal{O}(m^2)$
- converge in m iteration towards the solution α such that $K\alpha = \mathbf{y}$
- can be extended to non linear equations
- polynomial and subspace interpretation

$$\alpha_k = \mathbf{q}_{k-1}(\mathbf{K})\mathbf{y} \quad \mathbf{r}_k = p_k(K)\mathbf{y}$$

$$\mathbf{r}_k = \mathbf{y} - K\alpha_k \implies$$

$$p_k(K) = I - Kq_{k-1}(K)$$

$$\alpha_k \in \text{span}\{\mathbf{y}, K\mathbf{y}, K^2\mathbf{y}, \dots, K^{k-1}\mathbf{y}\}$$

$$\alpha_0 = 0$$

$$\alpha_1 = a_1 \mathbf{y}$$

$$\alpha_2 = a_1 \mathbf{y} + a_2 \left(\underbrace{\mathbf{r}_1}_{\mathbf{y} - a_1 K \mathbf{y}} + b_1 \mathbf{y} \right)$$

$$= q_1 \mathbf{y} + q_2 K \mathbf{y}$$

4 conjugate gradient type methods

	CG	MR	CGNE	MR II
cost	$K\alpha = \mathbf{y}$ $\frac{1}{2}\alpha^\top K\alpha - \alpha^\top \mathbf{y}$	$K\alpha = \mathbf{y}$ $\ K\alpha - \mathbf{y}\ ^2$	$K^2\alpha = K\mathbf{y}$ $\frac{1}{2}\alpha^\top K^2\alpha - \alpha^\top K\mathbf{y}$	$K\alpha = \mathbf{y}$ $\ K\alpha - \mathbf{y}\ ^2$
conj.	$\mathbf{d}_k^\top K \mathbf{d}_j = 0$	$\mathbf{d}_k^\top K^2 \mathbf{d}_j = 0$	$\mathbf{d}_k^\top K^2 \mathbf{d}_j = 0$	$\mathbf{d}_k^\top K^2 \mathbf{d}_j = 0$
$\mathbf{r}_k$	$\mathbf{y} - K\alpha$ $\mathbf{r}_k^\top \mathbf{r}_j$	$\mathbf{y} - K\alpha$ $\mathbf{r}_k^\top K \mathbf{r}_j$	$K\mathbf{y} - K^2\alpha$ $\mathbf{r}'_k{}^\top K^2 \mathbf{r}'_j$	Lanczos

summary of the definition of some Krylov subspace algorithms - conjugate gradient (CG), minimal residual (MR), conjugate gradient on normal equation (CGNE) and minimal residual for **non positive matrices (MR II)**.

Conclusion

Use iterative algorithm and early stopping to regularize :

- it's fast
- it's generic
- it regularizes
- it's accurate : good empirical results
- **strong link with PSL**

let's prove something - semiconvergence